Why Compute Return Map Derivatives?

- We want to compute periodic Rossler orbits that close after one return, two returns, or more to the vertical section $y_1 = c$.
- The parameterization $(u_1, u_2) \mapsto (c, u_1, u_2)$ has two independent unknowns. Matching two returned coordinates to starting coordinates gives two equations.
- **Newton's Method** use derivatives of the returned coordinates wrt u_1 and u_2 to iteratively solve these equations. This deck computes derivatives for one return.
- For higher winding numbers build a block system or (not such a good idea) apply the chain rule to compositions of the first return map.

Functions of Two Section Coordinates

- Write $\boldsymbol{u} = (u_1, u_2)$. The **Parameterization** is

$$\boldsymbol{\psi} : \mathbb{R}^2 \to \mathbb{R}^3, \qquad \boldsymbol{\psi}(u_1, u_2) = (c, u_1, u_2)$$

- The **First Return Map** sends the starting coordinates to the next crossing coordinates, using the same crossing direction.

$$\boldsymbol{p} : U \subset \mathbb{R}^2 \to \mathbb{R}^2, \qquad \boldsymbol{p}(u_1, u_2) = \big(p_1(u_1, u_2), p_2(u_1, u_2)\big)$$

  The returned state is $(c, p_1(u_1, u_2), p_2(u_1, u_2))$.

- The **First Return Time** is a scalar function

$$\tau : U \to (0, \infty), \qquad (u_1, u_2) \longmapsto \tau(u_1, u_2)$$

  $U \subset \mathbb{R}^2$ is an open set of start coordinates with a smooth first return.

Rössler System and Crossing Direction

- The singularity in the hill-top section $y_3 = b/(c - y_1)$ is the new simplified section.

$$\frac{\partial \boldsymbol{y}}{\partial t} = \boldsymbol{f}(\boldsymbol{y}) = \begin{bmatrix} -y_2 - y_3 \\ y_1 + a y_2 \\ b + y_3(y_1 - c) \end{bmatrix}, \qquad \boldsymbol{y}(0, \boldsymbol{u}) = (c, u_1, u_2)$$

- The **Poincaré Section** and its negative-crossing part are

$$\Sigma = \{\boldsymbol{y} : y_1 = c\}, \qquad \Sigma^- = \{\boldsymbol{y} \in \Sigma : -y_2 - y_3 < 0\}$$

- Choose $U \subset \{(u_1, u_2) : u_1 + u_2 > 0\}$. Start on Σ^- and return to Σ^-.

$$\tau(\boldsymbol{u}) = \min\{t > 0 : y_1(t, \boldsymbol{u}) = c, \quad -y_2(t, \boldsymbol{u}) - y_3(t, \boldsymbol{u}) < 0\}$$

Return Map and Its Jacobian

- The definitions give the three identities

$$y_1(\tau(\boldsymbol{u}), \boldsymbol{u}) = c, \qquad p_1(\boldsymbol{u}) = y_2(\tau(\boldsymbol{u}), \boldsymbol{u}), \qquad p_2(\boldsymbol{u}) = y_3(\tau(\boldsymbol{u}), \boldsymbol{u})$$

- Each scalar p_i has a gradient. The **Jacobian Matrix** of the vector map $\boldsymbol{p}$ is

$$D\boldsymbol{p}(\boldsymbol{u}) = \begin{bmatrix} \dfrac{\partial p_1}{\partial u_1} & \dfrac{\partial p_1}{\partial u_2} \\ \dfrac{\partial p_2}{\partial u_1} & \dfrac{\partial p_2}{\partial u_2} \end{bmatrix}, \qquad \boldsymbol{p}(\boldsymbol{u} + \delta\boldsymbol{u}) \approx \boldsymbol{p}(\boldsymbol{u}) + D\boldsymbol{p}(\boldsymbol{u})\delta\boldsymbol{u}$$

- The chain rule includes the effects on trajectory and τ of changing u_1 and/or u_2.

Compute Derivatives at Fixed Time

- Define $S_{ik}(t, \boldsymbol{u}) = \partial y_i(t, \boldsymbol{u})/\partial u_k$ with t fixed, for $i = 1, 2, 3$ and $k = 1, 2$. Differentiating the initial condition gives

$$\boldsymbol{S}(0, \boldsymbol{u}) = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- Differentiate $\partial y_i/\partial t = f_i(\boldsymbol{y})$ using the chain rule and integrate with the state y.

$$\frac{dS_{ik}}{dt} = \sum_{r=1}^{3} \frac{\partial f_i}{\partial y_r} S_{rk}, \qquad \frac{d\boldsymbol{S}}{dt} = \begin{bmatrix} 0 & -1 & -1 \\ 1 & a & 0 \\ y_3 & 0 & y_1 - c \end{bmatrix} \boldsymbol{S}$$

- These sensitivities compare states at equal times. Evaluate $\boldsymbol{S}$ at $t = \tau(\boldsymbol{u})$ only after integrating.

Calc 3 Chain Rule at the Return

- For any scalar function $g(t, u_1, u_2)$, the chain rule gives

$$\frac{\partial}{\partial u_k} g(\tau(u_1, u_2), u_1, u_2) = \frac{\partial g}{\partial t}\frac{\partial \tau}{\partial u_k} + \frac{\partial g}{\partial u_1}\frac{\partial u_1}{\partial u_k} + \frac{\partial g}{\partial u_2}\frac{\partial u_2}{\partial u_k}$$

  For $k = 1$ the last two factors are $1, 0$. For $k = 2$ they are $0, 1$.

- Apply this to $g = y_i$. Write $\tau_k = \partial\tau/\partial u_k$ and $p_i = p_i(\boldsymbol{u})$.

$$\boxed{\frac{\partial}{\partial u_k} y_i(\tau(\boldsymbol{u}), \boldsymbol{u}) = f_i(c, p_1, p_2)\tau_k + S_{ik}(\tau(\boldsymbol{u}), \boldsymbol{u})}$$

- $f_i(c, p_1, p_2)\tau_k$ accounts for the changing return time. S_{ik} accounts for the changing initial condition at fixed time. Everything is evaluated at $t = \tau(\boldsymbol{u})$.

Differentiate the Return Time

- The section is $y_1(\tau(\boldsymbol{u}), \boldsymbol{u}) = c$. Since c is fixed, the chain rule gives, for $k = 1, 2$,

$$0 = \big(-p_1(\boldsymbol{u}) - p_2(\boldsymbol{u})\big)\tau_k + S_{1k}, \qquad -p_1(\boldsymbol{u}) - p_2(\boldsymbol{u}) < 0$$

- Solve explicitly for the two partial derivatives.

$$\boxed{\frac{\partial \tau}{\partial u_1} = \frac{S_{11}}{p_1 + p_2}}, \qquad \boxed{\frac{\partial \tau}{\partial u_2} = \frac{S_{12}}{p_1 + p_2}}$$

- Here $S_{ik} = S_{ik}(\tau(\boldsymbol{u}), \boldsymbol{u})$ and $p_i = p_i(\boldsymbol{u})$. The chosen crossing gives $p_1 + p_2 > 0$. A small denominator can make the return derivatives large.

Differentiate Both Return Coordinates

- Apply the same chain rule to $p_1 = y_2(\tau, \boldsymbol{u})$ and $p_2 = y_3(\tau, \boldsymbol{u})$. At the return, $f_2(c, p_1, p_2) = c + ap_1$ and $f_3(c, p_1, p_2) = b$.

$$\frac{\partial p_1}{\partial u_k} = S_{2k} + (c + ap_1)\tau_k, \qquad \frac{\partial p_2}{\partial u_k} = S_{3k} + b\tau_k$$

- Substitute $\tau_k = S_{1k}/(p_1 + p_2)$ to obtain all four entries.

$$D\boldsymbol{p}(u_1, u_2) = \begin{bmatrix} S_{21} + \dfrac{c + ap_1}{p_1 + p_2} S_{11} & S_{22} + \dfrac{c + ap_1}{p_1 + p_2} S_{12} \\ S_{31} + \dfrac{b}{p_1 + p_2} S_{11} & S_{32} + \dfrac{b}{p_1 + p_2} S_{12} \end{bmatrix}$$

- Interpretation:
  - Row i contains the partial derivatives of p_i.
  - Column k describes changes in the starting coordinate u_k.

Finite Difference Verification

1. Specify a, b, c and $(u_1, u_2) \in U$. Initialize $\boldsymbol{y} = (c, u_1, u_2)$ and $\boldsymbol{S} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$.
2. Integrate the state equations and sensitivity equations. Locate the first ($\tau > 0$) negative crossing of $y_1 - c = 0$.
3. At the located event construct $\nabla\tau$ and $D\boldsymbol{p}$.
4. For ε small (but not too small) the centered differences

$$D\boldsymbol{p}(\boldsymbol{u})\boldsymbol{e}_k \approx \frac{\boldsymbol{p}(\boldsymbol{u} + \varepsilon\boldsymbol{e}_k) - \boldsymbol{p}(\boldsymbol{u} - \varepsilon\boldsymbol{e}_k)}{2\varepsilon}$$

where $\boldsymbol{e}_1 = (1, 0)$ perturbs in u_1 and $\boldsymbol{e}_2 = (0, 1)$ in u_2.